

Lesson Four Overview

Throughout the lecture, we are giving you some “*food for thought*”-- these are questions we suggest you think about as you go through this lesson. You do not need to submit solutions.

Learning Objectives

Students will be able to:

- See examples of real networks with highly skewed degree distributions
- Understand the math of power-law distributions and the concept of “scale-free” networks
- Learn about models that can generate networks with power-law degree distribution
- Explain the practical significance of power-law degree distributions through case studies

Required and Recommended Reading

Required Reading

- **Chapter 4 (sections 4.1, 4.2, 4.3, 4.4., 4.7, 4.8, 4.12)**, A-L. Barabási , [Network Science](http://www.networksciencebook.com/)  (<http://www.networksciencebook.com/>), available online, 2015.
- **Chapter 5 (sections 5.1, 5.2, 5.3)**, A-L. Barabási , [Network Science](http://www.networksciencebook.com/)  (<http://www.networksciencebook.com/>), available online, 2015.

Assignments

 L4: Knowledge Check

L4: Quiz